

# CE 336: Fluid Mechanics laboratory

**Fall 2025**

**Lecture 2**

By

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- ***MANDATORY:*** Students must complete “*Lab Safety Quiz*” online at:  
**Available on Canvas under the quizzes tab with 100% score**



[CSULB Home](#) > [COE](#) > [Lab Safety](#)

## Lab Safety

Wearing Personal Protection Equipment (PPE) is mandatory during face-to-face sections

- long pants and closed toed shoes.

1. Fluid Statics and Manometry\*

2. Free Jet experiment

3. Flow Measurement using Weirs.

4. Bernoulli's Experiment\*

5. Friction losses in Pipes

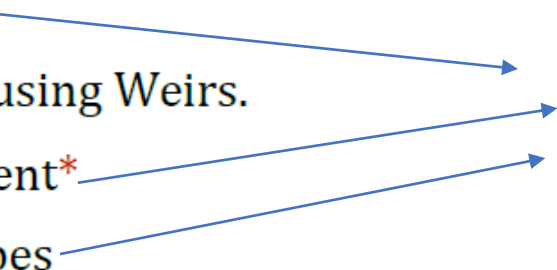
6. Minor losses in Pipe flow\*

7. Impulse (Pelton) Turbine

8. Centrifugal Pump characteristics

9. Hydraulic Jump in Open Channels

10. Wind tunnel experiment



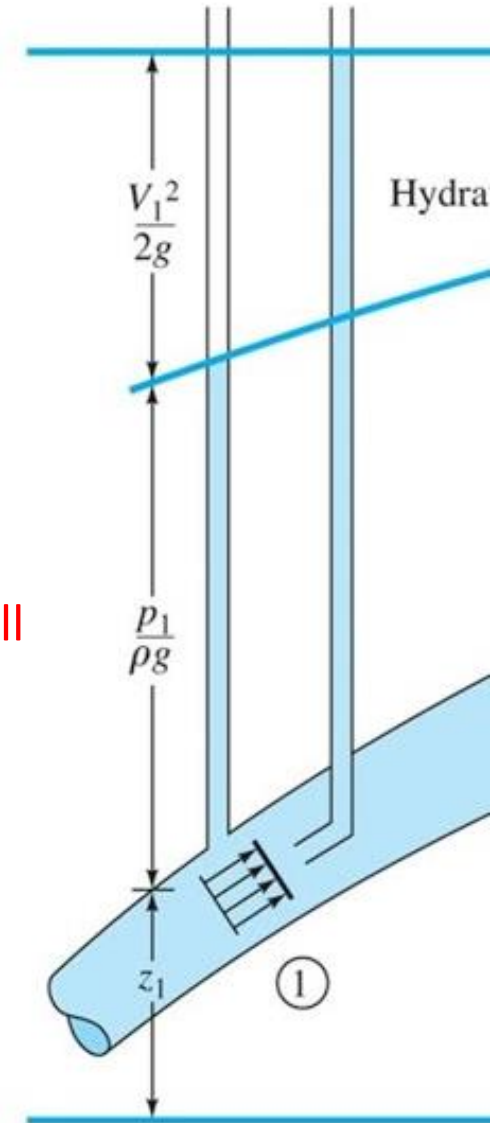
*3 first experiments*

## 3.4 Energy in Pipe Flow

The major portion of the energy is contained in three basic forms:

1. potential energy,
2. kinetic energy, and
3. pressure energy.

If we connect a monometer to a wall of pipe, the water rise shows the pressure energy

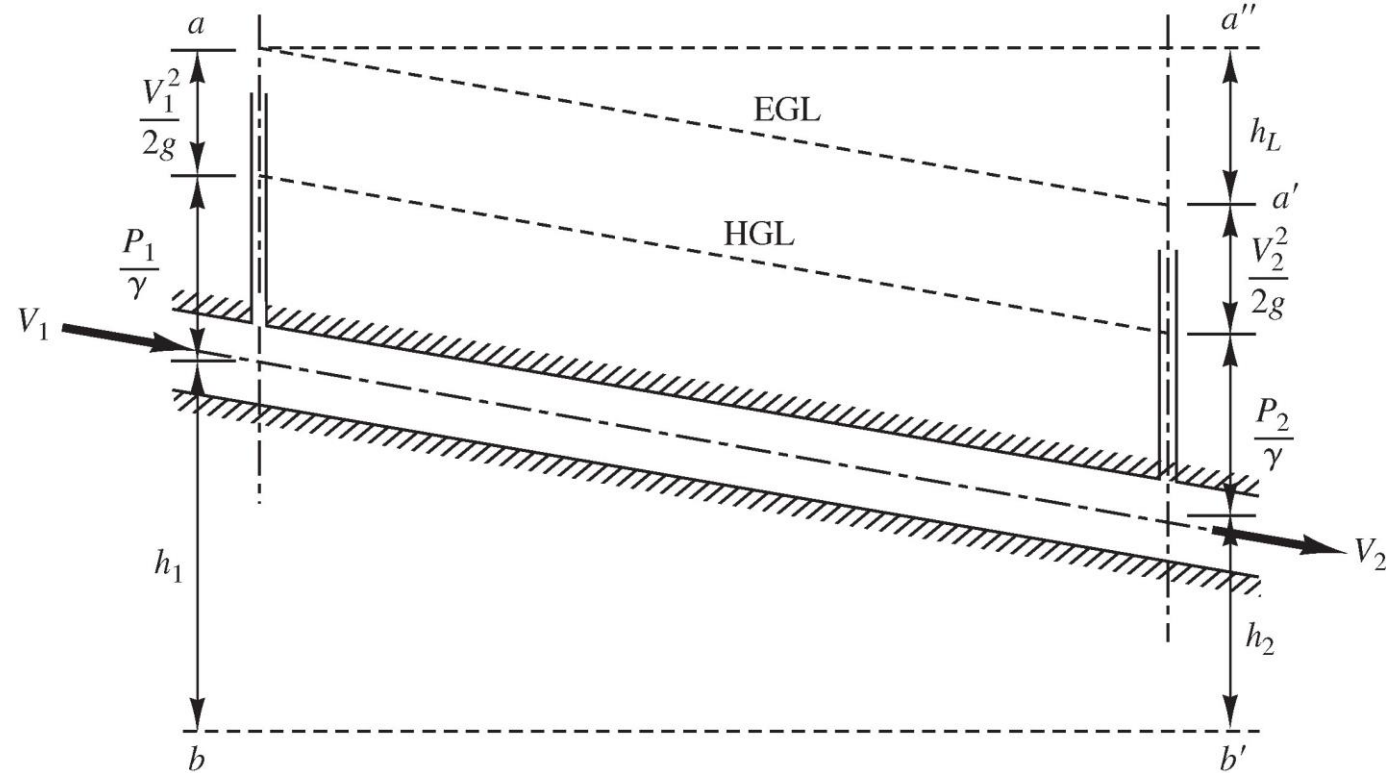


If we connect a monometer to the center of pipe, the water rise shows the total energy including velocity head.

## 3.4 Energy in Pipe Flow

The major portion of the energy is contained in three basic forms:

1. potential energy,
2. kinetic energy, and
3. pressure energy.



**Figure 3.5** Total energy and head loss in pipe flow

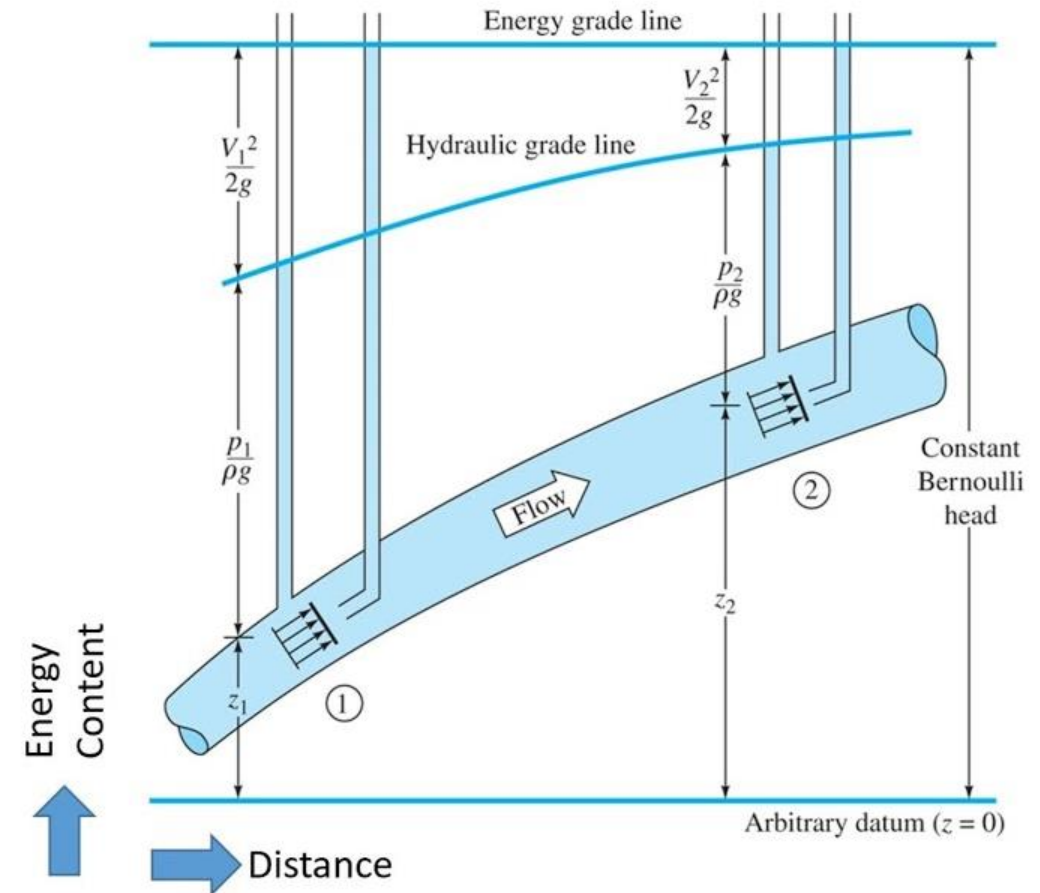
## Bernoulli's Equation Without energy loss

energy per unit weight of water

$$h_1 + \frac{P_1}{\gamma} + \frac{V_1^2}{2g} = h_2 + \frac{P_2}{\gamma} + \frac{V_2^2}{2g}$$

Elevation head      Pressure head      Velocity head

However, in reality  
energy loss occurs.





The algebraic sum of these three heads gives the point a above the energy datum.

between points a and b represents the total head,

Total head

EGL

$$H_1 = h_1 + \frac{P_1}{\gamma} + \frac{V_1^2}{2g}$$

Hydraulic grade line

Hydraulic grade

line

$$h_1 + \frac{P_1}{\gamma}$$

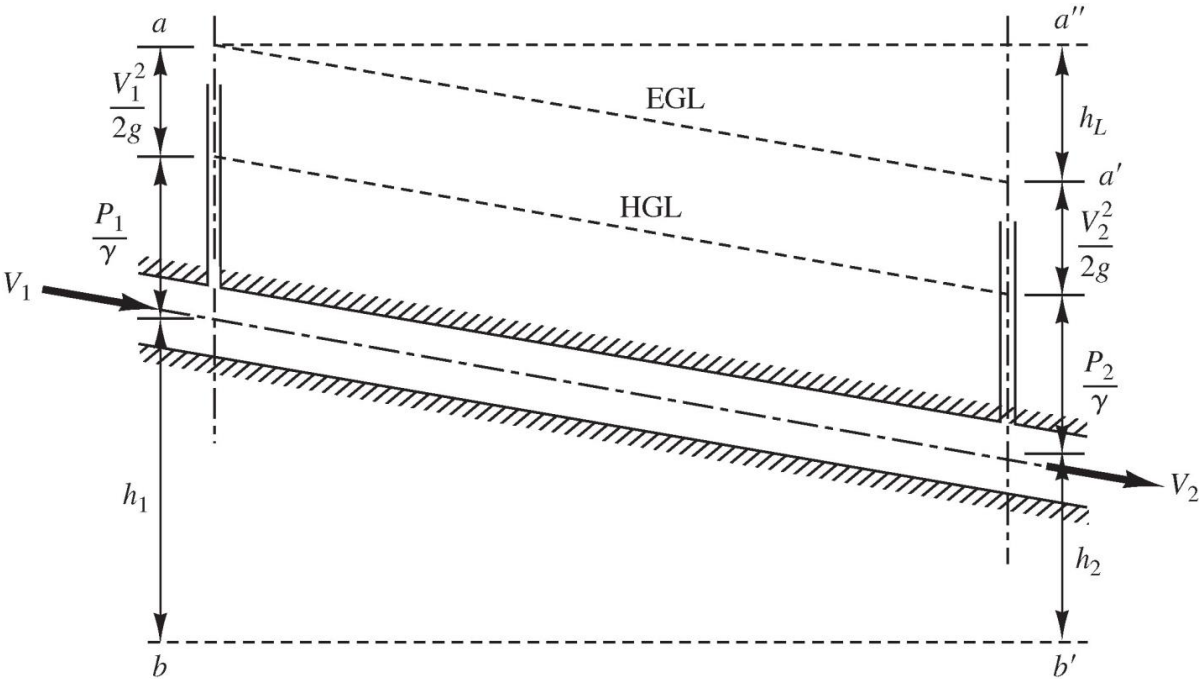
between the upstream and downstream sections,  
a certain amount of hydraulic energy is lost  
of friction (i.e., primarily converted to heat).

The elevation difference between points a' and a'' represents the head loss (h<sub>L</sub>)

Energy equation

$$h_1 + \frac{P_1}{\gamma} + \frac{V_1^2}{2g} = h_2 + \frac{P_2}{\gamma} + \frac{V_2^2}{2g} + h_L$$

Head loss



energy grade line (EGL)

hydraulic grade line (HGL)

The slope of the EGL represents the rate at which energy is being lost along the pipe.

Energy equation

$$h_1 + \frac{P_1}{\gamma} + \frac{V_1^2}{2g} = h_2 + \frac{P_2}{\gamma} + \frac{V_2^2}{2g} + h_L$$

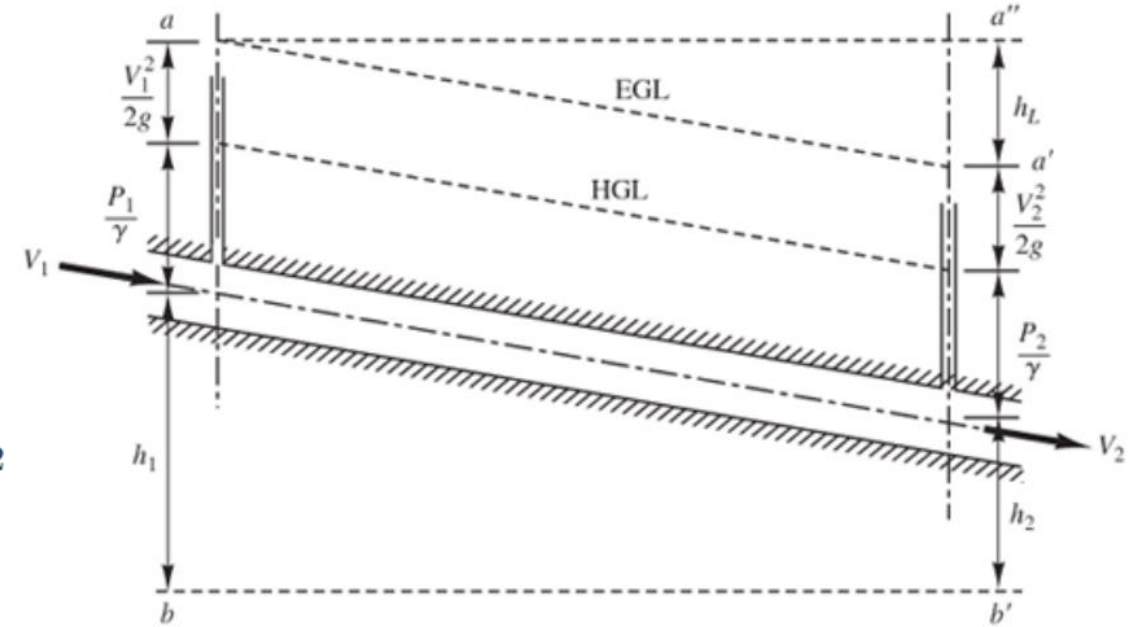
Head loss

- For a horizontal pipe of uniform size,

Pressure drop  $\frac{P_1 - P_2}{\gamma} = h_L$

- For a pipe with a uniform size but different elevations  $h_1$  and  $h_2$

$$\frac{P_1 - P_2}{\gamma} = h_L + h_2 - h_1$$



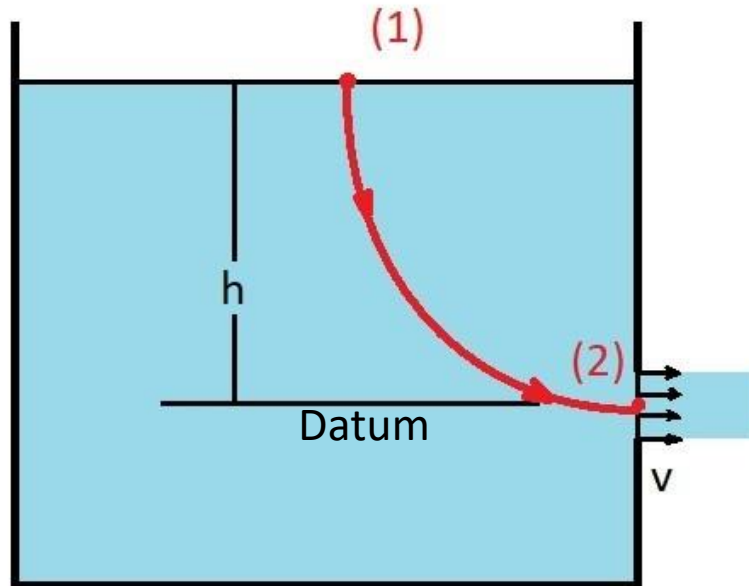
**Figure 3.5**  
Total energy and head loss in pipe flow



# Free jet, open tanks

Bernoulli's Equation  
Without energy loss

Find the velocity at outlet.



$$V_1=0$$

$$h_1 + \frac{P_1}{\gamma} + \frac{V_1^2}{2g} = h_2 + \frac{P_2}{\gamma} + \frac{V_2^2}{2g} + h_L$$

Point 1 and point 2 exposed to atmospheric pressure



$$P_1 = P_2 = 0$$

$$h = \frac{V_2^2}{2g}$$

$$V = V_2 = \sqrt{2gh}$$

In free jet example, we assume the size of tank is very large so the elevation of water in the tank does not vary.

## Determining coefficient of contraction, velocity and discharge

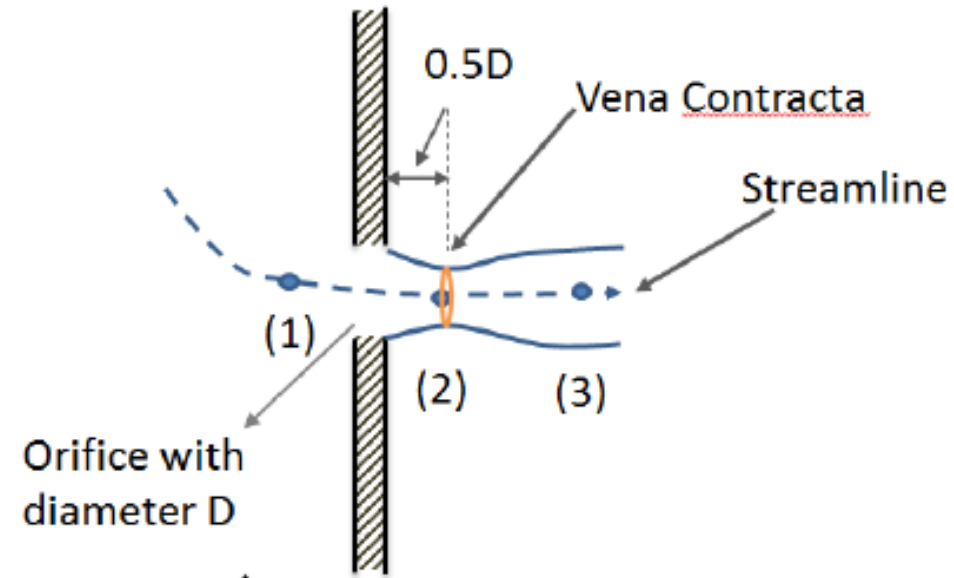


Figure 1. This reduced cross-sectional area is known as vena contracta (area at point (2) in Figure 1) and is approximately  $0.5D$  distance away from the orifice plate. Therefore,

$$A_c = C_c A_o \quad (1)$$

where  $C_c$  is the coefficient of contraction ( $C_c < 1$ ) and theoretically  $C_c = \frac{\pi}{\pi + 2} = 0.611$ . But,

$$A_c = C_c A_o$$

Theoretical velocity

$$V_t = \sqrt{2gh}$$

Actual velocity

$$V_a = C_v V_t$$

where  $C_v$  is the coefficient of velocity ( $C_v < 1$ ).

(~ 0.96 to 0.99). Actual flowrate  $Q_a$  through the orifice is the product of actual velocity at the vena contracta and area of the jet at the vena contracta.

$$Q_a = V_a A_c = C_v \sqrt{2gh} \times C_c A_o = C_v C_c A_o \sqrt{2gh} \quad (4)$$

$C_c$  and  $C_v$  can be combined to form coefficient of discharge  $C_d$ , such that  $C_d = C_c C_v$ . Therefore, Equation (4) can be written as:

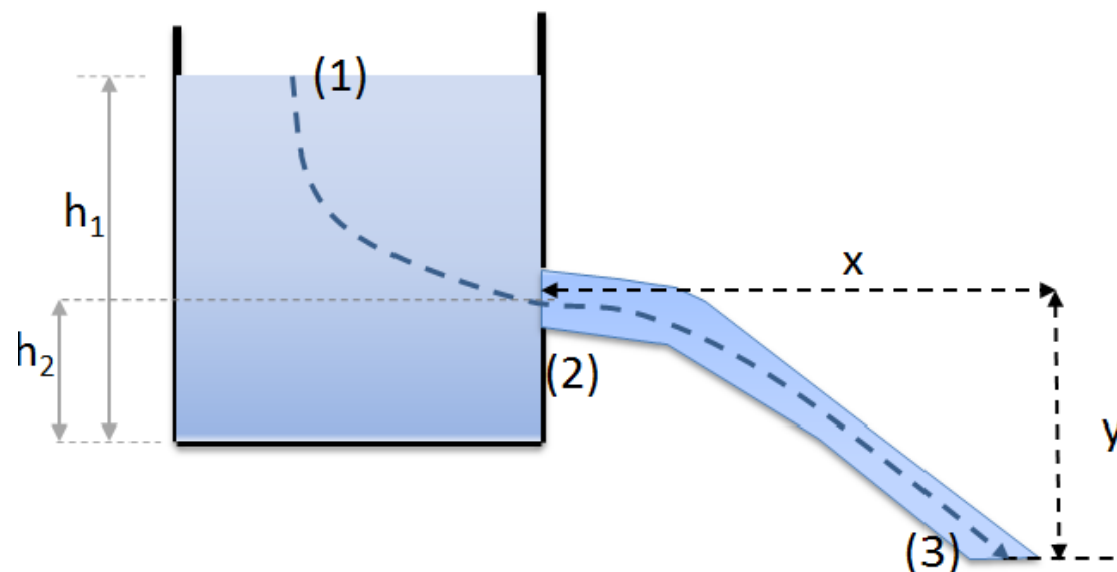
$$Q_a = C_d A_o \sqrt{2gh} \quad (5)$$

(Figure 2). The time  $t$  the jet takes to travel from point (2) to point (3) is equal to  $x/V_a$ . And the vertical distance  $y$  travelled by the jet is equal to  $gt^2/2$ . So, by measuring  $x$  and  $y$ , actual velocity can be determined. Knowing  $V_a$ , and  $V_t$  (Equation (2)),  $C_v$  can be determined as:

$$V_a = x / t$$

$$y = gt^2 / 2$$

$$C_v = \sqrt{\frac{x^2}{4yh}} = \frac{x}{2\sqrt{yh}}$$



## Calculations

### Exercise 1: Determining experimental $C_v$ , $C_c$ , $C_d$

# Theoretical time to empty the tank

$$t = \frac{2A_T}{C_d A_o \sqrt{2g}} [\sqrt{h_1} - \sqrt{h_2}]$$

## Discussions

1. Discuss if  $C_C$ ,  $C_v$ , and  $C_d$  is constant with head  $h$ .
2. Discuss if  $C_{d1}$  and  $C_{d2}$  are the same and if not comment how the coefficients are varying with  $h$ .
3. Comment the relationship observed between  $Q$  and  $h$ .
4. Comment on how kinetic energy efflux varies with  $h$ .
5. Comment on percent of error in time to empty the tank.

| No. of<br>Obs. | Head | X    | Y    | Vena-Contracta | Weight | Time  | Remarks |
|----------------|------|------|------|----------------|--------|-------|---------|
|                | (in) | (in) | (in) | (in)           | (lb)   | (sec) |         |
| 1              |      |      |      |                |        |       |         |
| 2              |      |      |      |                |        |       |         |
| 3              |      |      |      |                |        |       |         |
| 4              |      |      |      |                |        |       |         |
| <i>Σ</i>       |      |      |      |                |        |       |         |





